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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester M. Tech (CST) (CBCS) Degree Examinations

May- June 2025

22CST201 – ADVANCED DESIGN OF STEEL STRUCTURES

Duration: 3 Hours

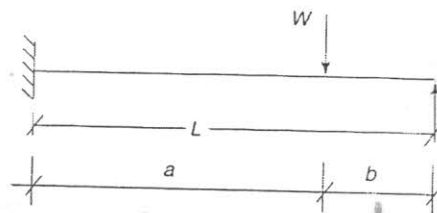
Max. Marks: 100

Note: Answer Five full questions choosing One full question from each Unit
IS:800-2007, IS: 811- 1987, IS 1161-1998 & IS:806-1968.

Unit – I

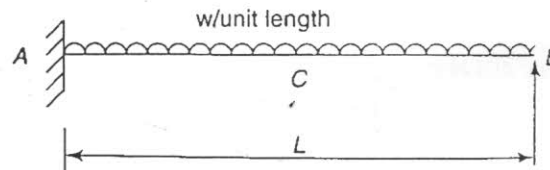
1. a) Explain with a neat sketch a stress vs. strain relationship of mild steel material.
- b) Determine the collapse load of the propped cantilever beam shown in fig.1 using kinematic method.

Marks	BT*	CO*	PO*
10	L*2	01	1,2

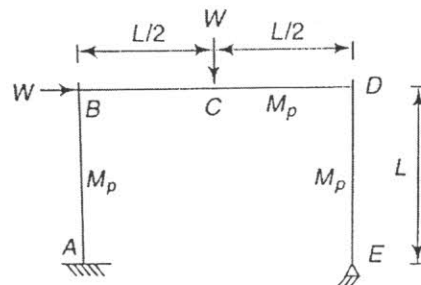


10	L2	01	1,2
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2. a) Determine the collapse load of the propped cantilever beam shown in fig.2.



- b) Determine the collapse load for the portal frame shown in fig.3 assuming that the beams and columns have the same cross section.



10	L2	01	1,2
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Unit – II

3. Design a laterally unrestrained beam to carry a UDL of 50kN/m. The beam is unsupported for a length of 1.5m and is simply placed on longitudinal beams at its ends.
4. Design a continuous beam of span 4.9m, 6m and 4.9m carrying a total UDL of 32.5 KN/m and laterally unrestrained with a bearing length of 100mm

20	L4	02	1,2
20	L4	02	1,2

P.T.O



Unit – III

5. A non-sway column in a building frame with a flexible joint is 4m high and subjected to the following load and moment:
Factored axial load = 500kN
Factored moment at top of column = 27kN-m
Factored moment at bottom of column = 45kN-m
Design a suitable beam column assuming $F_y = 250\text{N/mm}^2$ and take effective length $KL = 0.8L$. 20 L4 03 1,2
6. Design a beam column of length 3.75m it carries a compressive load of 500kN, a major axis moment of 5kN-m and minor axis moment of 2kN-m. Assume that effective length factor is 1.2 and the column is free to buckle in any plane. 20 L4 03 1,2

Unit – IV

7. a) What is castellated beam? Explain with a neat sketch. 8 L2 04 1,2
b) List and explain the possible failure modes of castellated beams. 6 L2 04 1,2
c) Describe the design procedures of castellated beam. 6 L2 04 1,2
8. Design a castellated beam in grade Fe-410 steel to carry an imposed load of 4kN/m and dead load of 4kN/m over a simply supported span of 16m. Assume that the compression flange is fully restrained. 20 L4 04 1,2

Unit – V

9. a) State the advantages and disadvantages of tubular steel structures. 10 L2 05 1,2
b) A tubular steel column of 7.8m length is hinged at both ends. It has nominal diameter of 225mm and Y_{st} 25 grade. Determine the safe load carrying capacity of column. 10 L4 05 1,2
10. A tension member carrying a force of 40kN meets the principal tie of tubular truss at an angle of 45° . If the force in principal tie is 100kN. Design the members and welded joint between the two tubes. Use Y_{st} 22 grade of steel. Assume their member are placed around inside location. 20 L4 05 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
